

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY  
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Fall 2023  
Data Structures and Algorithms (CS-212T)  
Assignment # 1

Semester: 3<sup>rd</sup>  
Announced Date: 01-11-2023  
Total Marks: 4  
Instructor Name: Adeel Saeed

Batch: 2022F  
Due Date: 08-11-2023  
Marks Obtained:

| CLO # | Course Learning Outcomes (CLOs)                                                     | PLO Mapping       | Bloom’s Taxonomy      |
|-------|-------------------------------------------------------------------------------------|-------------------|-----------------------|
| CLO 1 | Understand fundamental data structures as well as algorithms and their complexities | PLO_1 (Knowledge) | C2<br>(Understanding) |

Q1.What is the running time of the following algorithm (using Big-O notation). Show your analysis in a detailed analysis which counts the number of times each line of code executes.

```
i)    int sum = 0;
      for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
          for (k = 0; k < n; k++) {
            if (i == j == k) {
              for (l = 0; l < n*n*n; l++) {
                sum = i + j + k + l;
              }}}
        }
      }

ii)   int array_sum(int a, int n) {
      int i;
      int sum = 0;
      for (i = 0; i < n; i++) {
        sum = sum + a[i]
      }
      return sum;
    }
```

Q2. Show the series of steps taken by the Bubble Sort and Selection Sort algorithm while sorting following list in ascending order.

|       |   |    |   |    |    |   |    |    |
|-------|---|----|---|----|----|---|----|----|
| index | 0 | 1  | 2 | 3  | 4  | 5 | 6  | 7  |
| value | 9 | 20 | 6 | 10 | 14 | 8 | 60 | 11 |